CTL PROVISIONAL DECISION SHEET (PDSH)

Standard(s) (incl. year)	Subclause(s)	No.	Year
IEC 60950-1: 2005 IEC 60950-1: 2005/A1:2009 IEC 60950-1: 2005/A2:2013	4.3.6	1062	2013
Category			
OFF			
Subject	**Keywords**	**Developed by**	**To be approved**
Evaluation of interchangeable plug adapter design and potential misuse hazards - electrical connection	Interchangeable plug adapter	ETF2	2015 CTL Plenary Meeting

Question

For a power supply unit with several interchangeable plug adapters that fit socket-outlets for a wide range of countries, the electrical connection to the mains may be made via an IEC 60320-1 appliance inlet (C6, C8 type) as part of the power supply, and an IEC 60320-1 connector (C7, C5 type) as part of the interchangeable plug adapter. This leads to the following concerns and question:

1) As the coupling mechanism is utilizing a standard connector (C7, C5), how to avoid the potential misuse by users connecting the interchangeable plug adapter to other equipment which might be heavier and thus result in an excessive torque during the test of subclause 4.3.6?

2) When using the interchangeable plug adapter, the power supply becomes a direct plug-in equipment. The weight of the power supply is supported by the interchangeable plug adapter via the coupling mechanism when plugged into the wall socket-outlet. Is it permitted that the coupling relies solely on the friction provided by the appliance coupler (appliance inlet plus connector)?

Decision

1) Equipment investigated to IEC 60950-1 with interchangeable plug adapters shall be designed to meet the principles and requirements of the Standard.

   Typically, this can be accomplished by the following constructions and considerations. Other options are possible.

   a: Special shape of the coupling device (Type 1) where, for example, appliance couplers other than IEC 60320-1 standard sheets are used, or

   b: Special coupling mechanism system design (Type 2) where a misuse is unlikely due to the provided design, or

   General coupling mechanism system (Type 3) shall not to be accepted in general due to possible

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mechanical overload conditions of a wall socket outlet caused by misuse.

See page 3 for illustrations of Types 1, 2 and 3.

In accordance with the principles of subclauses 1.5.2 and 3.2.4, power supplies with interchangeable plug adapters designed into the equipment shall be subjected to the relevant mechanical durability (strength) tests specified in the applicable connector standards:
- IEC 60320-1 for standard connectors (Types 2), or
- IEC 61984, or other appropriate standard, for non-standard connectors (Type 1).

As required by subclauses 2.1.1 and 3.2.4, interchangeable plug adapters using a special shape of the coupling device (Type 1) are required to comply with all relevant accessibility requirements during all user handling conditions, including on the coupling side (opposite to the mains plug), while the power supply is detached.

2) No. The electrical connection of connector and appliance inlet shall not be affected by the weight of the power supply after assembly and plugging into the wall socket-outlet. This has to be verified for any position (orientation) of normal use.

Explanatory notes

2) a) Subclause 3.2.4 contains construction requirements in its third dashed paragraph. These apply generally and include above described equipment design where the appliance coupler is used as electrical connection of power supply and interchangeable plug adapter.
2) b) Excessive torque during the test of subclause 4.3.6 may have a negative effect on wall socket-outlets, resulting in subsequent hazards. IEC 60950-1 does not contain particular requirements for the consideration of misuse for such assemblies (i.e. misapplication of the interchangeable plug adapter). Markings and instructions ("instructional safeguards" in IEC 62368-1) are proposed to be sufficient to minimize risks possibly caused by misuse as described here.

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Attachment: examples of interchangeable plug adapters



Fig. 1: Plug adapter with special shape of the coupling device (e.g. appliance couplers other than IEC 60320-1 standard sheets) (Type 1)



Fig. 2: Plug adapter with special coupling mechanism system design but using IEC 60320-1 appliance coupler (Type 2)



Fig. 3: Plug adapter with general coupling mechanism system and using IEC 60320-1 appliance coupler (Type 3)

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